

Aaryan Samanta

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High school researcher applying machine learning to biology and medicine. First author of two peer-reviewed papers, USACO Platinum, and founder of AI Ethos, Inc., a 501(c)(3) that has reached more than 1,500 students with AI tutoring.

Education

Legend College Preparatory, Cupertino, CA

Class of 2028

GPA 4.8 weighted, 4.0 unweighted. SAT 1570 (800 Math). AP Scholar with Distinction: 9 exams, 4.77 average score.

NSHSS National Scholar. College Board National Recognition Program, Outstanding Academic Achievement.

Advanced AI sequence capstones: CNN pneumonia detector, symptom and disease pattern analysis, diabetes-risk model comparison ([code](#)).

Research and publications

First author, IEEE AIAM 2025

2025

Quantum-Inspired Hybrid Genetic Algorithm and Graph Neural Network Ensemble for Multimodal Classification. [IEEE Xplore](#)

Quantum-inspired genetic algorithm fused with a graph neural network, explained with SHAP. F1 0.57 vs 0.55 (Random Forest) and 0.52 (SVM).

First author, International Journal of High School Research

2026

Natural Genetic Variation in Mitochondrial Health-Regulating Genes in the *C. elegans* Strains CX11314 and EG4725 Leads to Increased Mitochondrial Resilience. [doi:10.36838/IJHSR816.1](https://doi.org/10.36838/IJHSR816.1)

UC Santa Barbara Summer Research Academy capstone (youngest accepted scholar). 540 isotypes, 609 RNA-seq samples, six mitochondrial quality-control genes; expression predictor Spearman 0.959.

Research Scholar, Stanford Medicine, Department of Urology

Jan 2026 to present

Mentor: Christos E. Constantinou, Associate Professor of Urology, Emeritus

Analyze anonymized pelvic-floor ultrasound from 23 subjects; extract displacement, velocity and acceleration during voluntary contraction, Valsalva and cough. Preliminary findings; not yet published.

Student Researcher, MIT EECS (mentored)

2026 to present

Weekly research with an MIT computer science professor applying machine learning to computational linguistics.

Researcher, UCLA COSMOS, Cluster 1: Brain-Inspired Computing

Summer 2026

InfernoCommand: actor-critic, fire-relative reinforcement learning for wildfire response on a real Los Angeles terrain model. Over training, mean reward rose 52% and buildings lost fell 54%. Manuscript submitted. [Code](#).

Manuscripts under review and in progress

2026

Under review: MS-TAGNet, a joint STFT-CWT multi-scale fusion framework for EEG semantic decoding under noisy conditions. **In progress (not yet accepted):** QIWGA-GAT for five-year tumor recurrence prediction, targeting IET Computer Vision; a study of how simplified biology diagrams create misconceptions, targeting *CBE-Life Sciences Education*.

Competitions

USACO Platinum division (2026). Perfect 1000 in Bronze, Silver and Gold, ranked first among 5,137, 2,721 and 1,336 participants. 2025 US Open: perfect 1000 in Gold and Silver, one of eight students in the U.S.

AMC / AIME National 1st place with perfect 150 scores: AMC 8 and 10 (2025); AMC 10 and 12 (2026). AIME qualifier in both years.

USAPhO Gold medalist (2026). Second-highest F=ma score in the U.S. (24 of 25).

USABO Semifinalist (2026). 43 of 50 on the Open Exam; Semifinal Recognition of Academic Excellence, top 50 of 573.

Kaggle Silver medal, Stanford RNA 3D Folding Part 2 (2026): 92nd of 1,867. [Competition](#).

ARML Local Team WARML I-5, 2nd of about 250 teams and 1st among U.S. teams (2026).

Leadership and service

Founder, President and CEO, AI Ethos, Inc. (501(c)(3))

2025 to present

Multilingual, privacy-first AI tutoring for low-income, rural, multilingual and neurodiverse students. More than 1,500 students reached; average 22% gain in math scores. IRS approval September 2025.

Founder and President, NextGenAI International

2025 to present

Student-led AI education organization: bootcamps, mentored projects and exchange, including outreach in Taiwan and Thailand. 200+ participants across three countries.

Founder and research lead, CogArc

In development

Wearable and EEG neurofeedback platform testing whether emotion is encoded in how body signals couple. Hypothesis stage.

Co-captain and student coach, Cupertino Junior Olympiad Math/CS Team

2024 to present

Coached 20+ peers toward AIME qualification and USACO Gold.

Volunteer team leader, VA Palo Alto Health Care System

2024 to present

Led a student team that audited medical images of 50+ elderly patients and standardized the workflow.

Peer tutoring and STEM outreach

Ongoing

Tutored 30+ peers in multivariable calculus and graph theory; mentored 20+ elementary students in coding.

Beyond the classroom

Track and field, Monta Vista Matadors

2025 to present

Freshman-year school record, 100 m (11.89); Rookie of the Year. Anchored the winning frosh/soph 4×400 m relay at the Los Gatos Top 8 Invitational (3:36.02); ran on the 4×400 m relay that set an all-time stadium record at RustBuster. 2026 season bests: 200 m 24.35, 400 m 54.53.

Percussion section leader, Monta Vista Concert Band

Grade 9

Led rehearsals for the first unanimous Superior rating across all Monta Vista ensembles at the CMEA Festival.

Skills

Languages and tools Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), Java, JavaScript, MATLAB, Git, LaTeX, Linux.

Spoken English; Spanish (AP level); Bengali (beginner).

References available on request.